Mathematical Set Operations Library

By Tanmay Vijay Sherkar / Sam-Altman

Index

1. Problem Statement
2. Your solution
3. Working (Method/Mind map)
4. Conclusion.

Note: Slides (excluding Thank-you and Introduction slide) should not be more than 4.

Problem Statement

- In many areas of computer science and data processing, it is important to manipulate and analyze collections of data efficiently. One common mathematical concept used for this purpose is **set operations**, which include **Union, Intersection, and Difference**. These operations help in identifying common elements, combining datasets, and finding unique elements between different collections.
- However, beginners often find it difficult to understand how these mathematical operations can be implemented programmatically using arrays and logical conditions. To bridge this gap, there is a need to develop a simple and interactive tool that demonstrates how set operations work on numeric data.
- This project aims to develop a **Mathematical Set Operations Library using JavaScript** that performs core set operations on multiple numeric arrays. The system will accept numeric arrays as input and dynamically compute results in the browser. The library will support essential set operations such as **Union, Intersection, and Difference**, allowing users to easily perform and visualize these operations.
- The main objective of this project is to strengthen students' understanding of **JavaScript arrays, logical problem-solving, and mathematical thinking**. By implementing these operations in JavaScript, students will gain foundational skills useful for **data processing, algorithm design, and backend development**.

My Solution

- Approach Used
- To solve the problem, I developed a **browser-based JavaScript application** that performs mathematical set operations on numeric arrays entered by the user.
- The application follows a **three-step process: input, processing, and output**.

1. User Input Handling

The user enters numeric arrays in a **comma-separated format** such as:

1, 2, 3, 4

The program then:

- Splits the text using commas
- Removes extra spaces
- Converts values into numeric arrays
- Filters out invalid inputs

This ensures the program only processes **valid numbers**.

2. Implementation of Set Operations

Three reusable JavaScript functions are used to perform the required set operations.

Union

Combines unique elements from both arrays.
Example:
[1,2,3] ∪ [3,4] → [1,2,3,4]

Duplicates are removed to maintain **set uniqueness**.

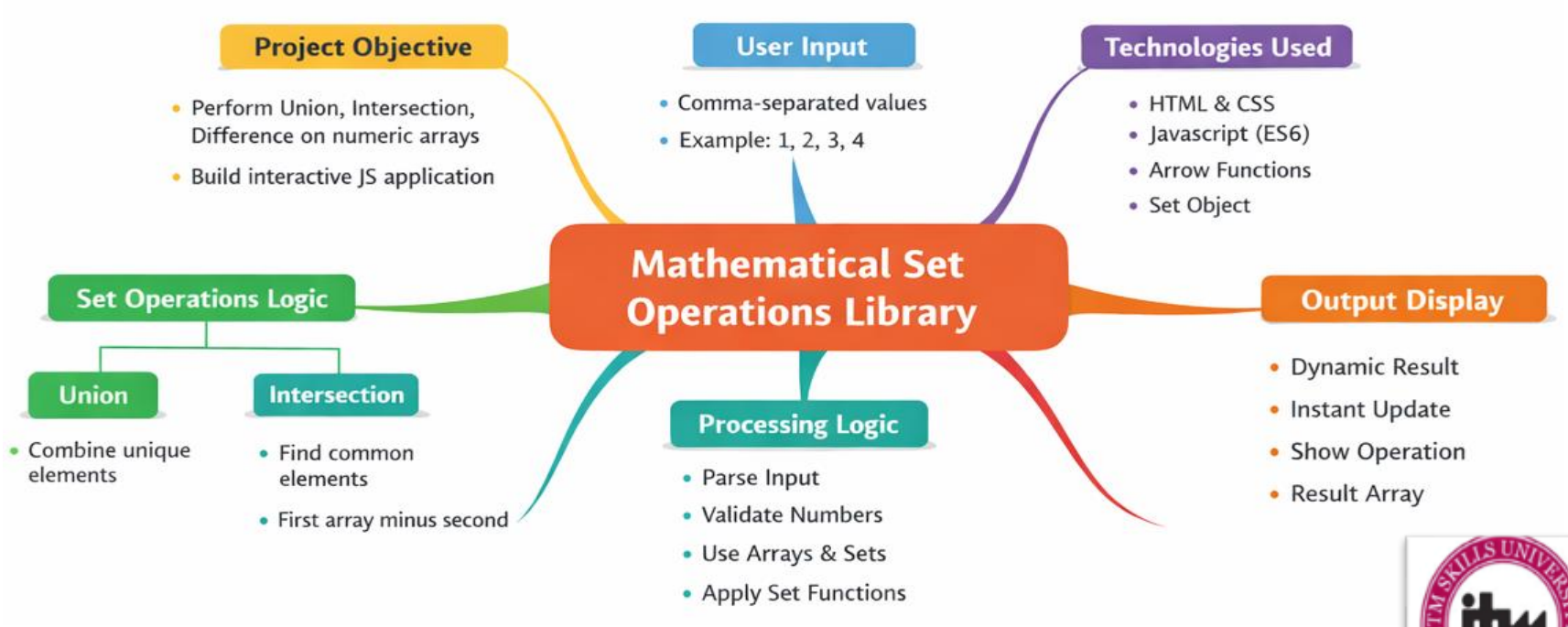
Intersection

Returns elements that are common in both arrays.
Example:
[1,2,3] ∩ [2,3,5] → [2,3]
The program compares elements of both arrays to identify **shared values**.

Difference

Returns elements that exist in the first array but not in the second.
Example:
[1,2,3] − [2] → [1,3]
This operation helps identify **unique elements of the first set**.

Working (Method/Mind map)



Conclusion

- The **Mathematical Set Operations Library** project successfully demonstrates the implementation of fundamental set theory concepts using modern web technologies. The application allows users to input numeric arrays and dynamically perform core set operations such as **Union, Intersection, and Difference** directly in the browser.
- Through this project, key programming concepts such as **array manipulation, logical comparison, input validation, and dynamic DOM interaction** were applied to process user data efficiently. The use of **JavaScript ES6 features**, including arrow functions, spread operators, and Set objects, helped create a clean, efficient, and maintainable implementation.
- Additionally, the project highlights how mathematical logic can be integrated with **interactive web interfaces** to build simple yet powerful data-processing tools. Overall, this project strengthened practical understanding of **JavaScript programming, algorithmic thinking, and web-based application development**, making it a valuable learning experience in the field of software development.

ITM SKILLS UNIVERSITY

Thank You